

Validation Source Provenance & Lineage Module (VSPLM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VALIDOS® — Validation Governance Architecture

Parent Standard: Validation Source Reliability Standard (VSRS)

Operational Layer: Validation Source Reliability Governance Layer

Category: AI & Interpretation

Subcategory: Validation Governance

Type: Parent Standard Module

Governed Space: Validation Source Provenance & Lineage

Version: 1.0

Status: Canonical · Open Module

Effective Date: 9 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · OBIDENITY® · INTEGROS®
· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Validation Source Provenance & Lineage

Canonical Definition

Validation Source Provenance & Lineage Module (VSPLM) defines the governance framework for reconstructing, representing, assessing, and maintaining the material origin history, evidentiary pathway, upstream dependencies, intermediary relationships, transformations, and lineage through which Validation Evidence progresses from its originating conditions to its use within a governed validation process.

It establishes the conditions under which an identified and attributed source relationship becomes a **traceable Validation Source Chain**.

Operational Role

VSPLM serves as the second internal governance space of the Validation Source Reliability Standard.

The preceding module asks:

Who or what is the source, and can the evidence be attributed to it?

VSPLM asks:

Where did the evidence originate, through which pathway did it reach validation, what happened to it along that pathway, and upon which upstream sources does it depend?

The governed progression is:

Identified & Attributed Source → Origin Establishment → Provenance Reconstruction → Lineage Mapping → Intermediary Identification → Transformation Mapping → Dependency Assessment → Traceable Validation Source Chain

VSPLM does not determine whether the source is ultimately reliable.

It establishes the provenance and lineage structure required for that later determination.

Module Operational Space

VSPLM governs:

- validation-source provenance,
- evidence origin,
- source origin,
- provenance depth,
- provenance continuity,
- source lineage,
- upstream-source identification,
- downstream-source relationships,
- source chains,
- source dependency,
- shared-source dependency,
- intermediary relationships,
- transformation pathways,
- evidence derivation,
- aggregation lineage,
- provenance gaps,
- lineage gaps,
- provenance conflicts,
- lineage conflicts,
- provenance uncertainty,
- provenance boundaries,
- provenance change,
- lineage change,
- and provenance traceability.

Module Function

VSPLM applies after material Validation Sources have been sufficiently identified and evidence attribution has been established or bounded.

Its function is to prevent:

- immediate evidence providers being mistaken for original sources,
- downstream reports concealing upstream evidence origins,
- multiple derived sources being treated as independent origins,
- shared datasets creating artificial source diversity,
- intermediary transformations remaining invisible,
- aggregated evidence losing its source structure,
- source chains being collapsed into single-source labels,
- missing provenance being interpreted as complete provenance,
- derived evidence being represented as primary evidence,
- historical source changes being overwritten,
- and source-based confidence being assigned without understanding material upstream dependencies.

Provenance and Lineage Are Related but Distinct

VALIDOS® distinguishes:

Provenance

from:

Lineage

Provenance asks:

What material history brought this evidence from origin to validation use?

Lineage asks:

Which upstream sources, derived sources, dependencies, and source relationships contributed to the evidence?

Together they create a reconstructable source structure.

Evidence Origin

Evidence origin represents the earliest materially relevant point at which the evidentiary content was:

- observed,
- measured,
- generated,
- recorded,
- produced,
- or otherwise established.

The first entity encountered by a validator is not necessarily the

evidence origin.

For example:

Website → Report → Research Paper → Dataset → Original Measurement

The website may be the immediate provider.

The measurement may be the relevant evidentiary origin.

Origin Precision

Not every validation requires identical origin precision.

Depending upon the Validation Purpose, Scope, Depth, consequence, and evidentiary role, origin may need to identify:

- an organization,
- a specific system,
- a specific instrument,
- a particular dataset,
- a model version,
- an operational event,
- a person,
- a laboratory,
- a repository,
- or another source instance.

VSPLM requires sufficient origin precision for the reliability question being governed.

Provenance Chain

A Validation Evidence provenance chain may conceptually include:

Origin → Generation → Recording → Processing → Transformation → Transmission → Aggregation → Storage → Retrieval → Validation Use

Not every evidence item will contain every stage.

VSPLM identifies the material stages relevant to source reliability.

Provenance Depth

The required depth of provenance reconstruction should be proportionate to the validation role.

Low-consequence validation may require limited provenance.

High-consequence validation may require substantially deeper reconstruction.

Factors affecting provenance depth may include:

- Validation Depth,
 - intended reliance,
 - consequence of error,
 - evidence criticality,
 - number of intermediaries,
 - degree of transformation,
 - source uncertainty,
 - and dependency complexity.
-

Provenance Continuity

Provenance continuity exists where the material evidentiary pathway can be followed without unexplained breaks that materially affect source reliability.

Continuity does not require every technical action to be documented.

It requires that the material source transitions remain sufficiently reconstructable.

Provenance Gap

A **Provenance Gap** exists where a material portion of the evidentiary pathway cannot be sufficiently established.

Examples include:

- unknown original dataset,
- missing measurement origin,
- unidentified intermediary,
- undocumented transformation,
- unknown historical source,
- missing model version,
- or unavailable source records.

A Provenance Gap does not automatically make evidence unusable.

It creates a source-reliability condition that must remain visible.

Critical Provenance Gap

A Provenance Gap may become critical where the missing information prevents meaningful assessment of:

- source identity,
 - source capability,
 - independence,
 - transformation integrity,
-

- evidence origin,
- shared dependency,
- or another material reliability condition.

Criticality must be determined relative to the validation role.

Source Lineage

Source lineage represents the relational structure connecting evidence to the sources from which it derives.

A simple lineage may be:

Measurement Instrument → Dataset → Validation Evidence

A complex lineage may be:

Sensor Network → Local Database → Aggregation System → External Data Provider → Analytical Model → Human Report → Validation Evidence

VSPLM makes these dependencies visible.

Upstream Source

An **Upstream Source** is a source upon which another source or evidence object materially depends.

Upstream sources may include:

- primary datasets,
- original measurements,
- external databases,
- prior models,
- sensor systems,
- research studies,
- operational records,
- or other evidence origins.

Upstream dependency can materially affect downstream Source Reliability.

Downstream Source

A **Downstream Source** receives, processes, transforms, interprets, aggregates, or republishes evidentiary content derived from one or more upstream sources.

Downstream sources may add legitimate value.

However, their existence does not create a new independent

evidentiary origin automatically.

Derived Source Lineage

Derived sources must remain connected to material upstream origins where those origins affect validation reliability.

VALIDOS® establishes:

New Presentation ≠ New Evidentiary Origin

and:

New Analysis ≠ Independent Underlying Evidence

A derived source may contribute independent interpretation while still depending upon non-independent underlying evidence.

Shared Source Dependency

Two or more apparently separate sources may share the same upstream origin.

For example:

Source A → Dataset X

Source B → Dataset X

Source C → Dataset X

A superficial source count would produce three sources.

Lineage analysis reveals one shared upstream evidentiary dependency.

VSPLM therefore establishes:

Source Multiplicity ≠ Source Independence

Dependency Concentration

A validation evidence basis may appear diverse while depending heavily upon one source.

VSPLM may identify **Dependency Concentration** where a significant portion of the evidence basis ultimately depends upon:

- one dataset,
 - one organization,
 - one model,
 - one measurement system,
 - one repository,
 - one methodology,
-

- or another shared upstream source.

Dependency concentration does not automatically make evidence unreliable.

It identifies a structural reliability condition.

Source Chain

A **Validation Source Chain** represents the sequence of materially relevant source relationships through which evidence progresses.

Source chains may be:

Direct

Derived

Intermediated

Aggregated

Multi-Source

or:

Hybrid

The chain structure should remain sufficiently visible for subsequent reliability assessment.

Intermediary Identification

An intermediary is an entity or system that materially handles evidence between its upstream origin and validation use.

Intermediaries may:

- transmit,
- translate,
- normalize,
- aggregate,
- filter,
- classify,
- summarize,
- store,
- reformat,
- or interpret evidence.

VSPLM identifies intermediaries and their position in the

provenance chain.

Intermediary Role

Each intermediary may perform a different function.

Examples include:

Transmission Intermediary

Storage Intermediary

Transformation Intermediary

Aggregation Intermediary

Interpretation Intermediary

Publication Intermediary

Distribution Intermediary Role identification helps later governance determine which intermediaries materially affect Source Reliability.

Transformation Pathway

Evidence may change between origin and validation use.

VSPLM maps material transformations such as:

- filtering,
- cleaning,
- normalization,
- aggregation,
- translation,
- anonymization,
- statistical processing,
- compression,
- summarization,
- model processing,
- classification,
- or interpretation.

VSPLM does not determine whether each transformation was methodologically correct.

It establishes that the transformation occurred and where it sits within the source chain.

Transformation Visibility

VALIDOS® establishes:

A transformed evidence object should not be represented as if it were identical to its original source evidence.

Where transformation materially changes evidentiary representation, the transformation must remain visible.

Aggregation Lineage

Aggregated evidence may combine multiple origins.

For example:

Dataset A + Dataset B + Dataset C → Aggregated Dataset D

VSPLM preserves the relationship between the aggregated object and its material contributing sources.

Where source-level separation is impossible, that limitation must remain explicit.

Lineage Branching

One evidence object may derive from multiple upstream sources.

Likewise, one upstream source may contribute to many downstream evidence objects.

VSPLM therefore permits branching lineage structures rather than forcing evidence into a simple linear chain.

Conceptually:

Source A + Source B → Derived Evidence C

and:

Source A → Evidence D + Evidence E This becomes particularly important for large AI, scientific, financial, and operational evidence ecosystems.

Circular Lineage

A source chain may sometimes contain circular dependencies.

For example:

Model A uses Dataset B

while:

Dataset B includes outputs previously generated by Model A.

Circular lineage may create hidden dependence, reinforcement, contamination, or false corroboration.

VSPLM requires material circular relationships to remain visible.

Provenance Conflict

A Provenance Conflict exists where available records provide materially incompatible accounts of the evidence pathway.

Examples include:

- conflicting origin claims,
- incompatible timestamps,
- inconsistent source records,
- disputed transformations,
- or contradictory intermediary histories.

VSPLM preserves the conflict rather than forcing a single provenance narrative without sufficient basis.

Lineage Conflict

A Lineage Conflict exists where the upstream dependency structure cannot be reconciled.

For example:

one source claims an independent dataset,

while another record indicates that the dataset derives from a shared upstream repository.

Material lineage conflicts must remain visible for later Source Reliability Determination.

Provenance Uncertainty

Complete provenance may not always be obtainable.

Where evidence origin or pathway can only be partially reconstructed, VSPLM may classify provenance conditions such as:

Provenance Established

Provenance Substantially Established

Provenance Partially Established

Provenance Uncertain

Provenance Unavailable These classifications describe provenance quality.

They do not determine Source Reliability by themselves.

Lineage Uncertainty

Likewise, lineage may be:

Lineage Established

Lineage Partially Established

Lineage Uncertain

or:

Lineage Unknown

The required certainty should remain proportionate to the validation role.

Provenance Boundary

VSPLM does not require infinite backward tracing.

Every provenance reconstruction must eventually establish a reasonable boundary.

A Provenance Boundary may be reached when:

- sufficient origin has been established,
- further upstream information is immaterial,
- additional reconstruction is impossible,
- or the required Validation Depth has been satisfied.

The boundary should be explicit where it materially affects reliance.

Provenance and Primary Evidence

Primary evidence may possess a shorter provenance chain.

However:

Primary Evidence ≠ Automatically Reliable Evidence

The originating instrument may be defective.

The observer may lack competence.

The source may be conflicted.

The recording mechanism may be compromised.

VSPLM establishes provenance.

Subsequent VSRS modules govern those other reliability dimensions.

Provenance and Derived Evidence

Derived evidence is not inherently inferior.

Derived evidence may provide:

- stronger analysis,
- useful aggregation,
- improved interpretation,
- cross-source comparison,
- or higher operational value.

The governance requirement is that material derivation remains visible.

Provenance and Source Independence

VSPLM provides critical input into later independence assessment.

Two sources cannot be confidently treated as independent until their material lineage relationships are understood.

Therefore:

Apparent Independence → Lineage Assessment → Actual Independence Assessment

This prevents duplicated upstream evidence from producing false validation confidence.

Provenance and Corroboration

Corroboration requires more than agreement.

If three sources agree because all three reproduce the same upstream dataset, they may represent:

Three Presentations

but:

One Evidentiary Origin

VSPLM exposes this distinction before corroboration is used within

Provenance Change

Provenance knowledge may change when:

- new origin information emerges,
- hidden intermediaries are discovered,
- transformations become known,
- source records are corrected,
- upstream dependencies are identified,
- or historical records become available.

Material changes should update the provenance structure without erasing previous records.

Lineage Change

Lineage may change because:

- a dataset is replaced,
- a model changes training sources,
- a provider changes upstream supplier,
- a system begins using new external data,
- or a source dependency is removed.

Material lineage changes may affect existing Source Reliability Determinations.

Provenance Continuity Across VALIDOS®

Provenance information should remain connected to the evidence throughout:

**Source Reliability → Validation Execution → Validation Determination
→ Validation State → Reliance**

This allows later users of the validation result to reconstruct the material evidentiary origins behind it.

Minimum Implementation Framework

1. Establish Evidence Origin

Identify the earliest materially relevant evidence origin that can reasonably be established.

2. Reconstruct Material Provenance

Map the material pathway from origin to validation use.

3. Identify Upstream and Downstream Sources

Determine the relevant source relationships and dependencies.

4. Identify Intermediaries

Record material entities or systems through which evidence passed.

5. Map Material Transformations

Identify transformations capable of affecting evidentiary representation.

6. Assess Shared Dependencies

Determine whether apparently separate sources rely upon common upstream origins.

7. Identify Provenance and Lineage Gaps

Record missing, uncertain, disputed, or unreconstructable portions of the source structure.

8. Establish Provenance Boundary

Determine where provenance reconstruction can reasonably stop for the governed validation role.

9. Preserve Provenance & Lineage Traceability

Maintain:

- Validation Evidence reference,
- source identity references,
- evidence origin,
- provenance chain,
- source lineage,
- upstream sources,
- downstream sources,
- intermediaries,
- transformations,
- shared dependencies,
- dependency concentration,
- provenance gaps,
- lineage gaps,
- conflicts,
- uncertainty,
- provenance boundary,
- changes,
- version,
- and sufficient lifecycle traceability.

Provenance & Lineage Outputs

VSPLM may produce:

- Validation Evidence Origin Record,
- Validation Source Provenance Record,
- Validation Source Chain,
- Source Lineage Map,
- Upstream Source Register,
- Downstream Source Relationship Record,
- Source Dependency Map,
- Shared Dependency Record,
- Dependency Concentration Record,
- Intermediary Map,
- Evidence Transformation Pathway,
- Aggregation Lineage Record,
- Circular Lineage Record,
- Provenance Gap Record,
- Lineage Gap Record,
- Provenance Conflict Record,
- Lineage Conflict Record,
- Provenance Boundary,
- Provenance Change Record,
- and Provenance & Lineage Determination.

Use Case 1 — AI Benchmark Evidence

Scenario

Five organizations publish benchmark results supporting the performance of an AI system.

Application

VSPLM reconstructs their source lineage.

All five organizations used benchmark datasets derived from the same original dataset.

Three also used the same evaluation framework.

Result

The validation process no longer interprets the five reports as five completely independent evidentiary origins.

Their shared dependencies become visible for subsequent independence and reliability assessment.

Use Case 2 — Healthcare Dataset

Scenario A healthcare AI validation relies upon a commercial dataset supplied by a data provider.

Application

VSPLM determines that the dataset combines:

- hospital records,
- laboratory data,
- historical public datasets,
- and transformed records from another commercial provider.

Several transformation stages occurred before delivery.

Result

The commercial provider remains an identifiable source, but the validation process now possesses a lineage map revealing the upstream origins and transformations upon which the dataset depends.

Use Case 3 — Internet Claim

Scenario

The same factual claim appears across dozens of online sources.

Application

VSPLM traces the available source chains.

Most sources ultimately derive from one original report.

Several others cite articles that themselves cite that same report.

Result

Apparent widespread corroboration becomes correctly represented as a highly repeated but strongly concentrated evidentiary lineage.

Architectural Position

Validation Source Provenance & Lineage is the second internal governance space of the **Validation Source Reliability Standard (VSRS)**.

The progression now becomes:

Validation Evidence → Source Identity & Attribution → Source Provenance & Lineage

VSIAM establishes:

Who or what contributed the evidence?

VSPLM establishes:

Where did that evidence originate, how did it reach validation, and upon what upstream sources does it depend?

The next governance stage can then ask:

Are those sources actually capable and competent to perform the evidentiary roles assigned to them?

The VSRS progression therefore continues: **Identity & Attribution** → **Provenance & Lineage** → **Capability & Competence** → **Independence & Conflict** → **Source Reliability Determination**

Foundational Principle

Source diversity cannot be established until the lineage behind the sources is understood.

Canonical Closing Statement

Validation Source Provenance & Lineage Module establishes the traceability structure through which VALIDOS® reconstructs the material origins, pathways, dependencies, intermediaries, transformations, and source relationships underlying Validation Evidence. By distinguishing immediate providers from evidentiary origins, source multiplicity from independent lineage, and transformed evidence from its upstream foundations, VSPLM prevents hidden dependencies and incomplete provenance from creating artificial source confidence while establishing the traceable source chain required for subsequent capability, independence, conflict, corroboration, and Source Reliability governance.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Previous module: Validation Source Identity & Attribution Module \(VSIAM\)](#)

→ [Next module: Validation Source Capability & Competence Module \(VSCCM\)](#).

Related Documents

→ [Validation Source Reliability Standard](#)

→ [About Validation Source Reliability Standard](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>